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Practical Fixed-Time Tracking Control for a Class of Strict-Feedback Nonlinear Systems: A Fixed-Time Disturbance Observer Approach

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***Abstract*—This article develops a practical fixed-time tracking control framework for a class of strict-feedback nonlinear systems subject to unknown external disturbances. A fixed-time disturbance observer (FxTDO) is first designed to reconstruct the disturbances together with their higher-order derivatives within a uniform settling time whose upper bound is independent of the initial conditions. To avoid the explosion of complexity arising from the repeated differentiation of virtual controllers, a fixed-time command filter (FxTCF) is incorporated into the recursive backstepping design. In contrast to conventional command filters with persistent filtering errors, the proposed FxTCF guarantees that the filtering error vanishes within a fixed time. A switching function is further embedded into the virtual control laws to ensure the required second-order differentiability and enable the effective implementation of the FxTCF without introducing additional error compensation dynamics. Moreover, a dynamic event-triggering mechanism is integrated into the control framework to reduce unnecessary transmissions while preserving the desired closed-loop performance. Theoretical analysis establishes the boundedness of all closed-loop signals, the practical fixed-time convergence of the tracking error to a prescribed residual set, and the exclusion of Zeno behavior. Simulation results for a flexible robotic manipulator demonstrate the effectiveness and advantages of the proposed control framework.**

***Index Terms*—Dynamic event-triggered control, fixed-time command-filter, fixed-time disturbance observer (FxTDO), practical fixed-time stability.**

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I. Introduction

NONLINEARITY is an inherent characteristic of most practical systems, and nonlinear behaviors cannot be ignored in the modeling and control of engineering systems, particularly when high accuracy and robustness are required, such as flexible robotic manipulators, power systems, and uncrewed aerial vehicles [1], [2], [3]. In recent years, improvements in control algorithms for nonlinear systems have attracted widespread attention [4], [5], [6], [7], [8]. The study in [9] formulated an adaptive tracking control methodology applicable to strict-feedback nonlinear systems characterized by full-state constraints. However, the control algorithm in [9] suffered from the "complexity explosion" issue caused by the recursive derivations of the virtual controller. Command-filter technique have been widely employed to address the explosion-of-complexity problem in conventional backstepping control [10], [11], [12], [13]. The study in [14] addressed the design of a finite-time adaptive tracking controller with a singularity-free command-filter for a single-link flexible-joint robotic manipulator. Nevertheless, the command-filter adopted in the referenced study [14] was limited to the first-order command structure. In [15], a second-order command-filter was designed, and its effectiveness was ensured by incorporating a switching function. However, the error compensation mechanism was still required in [15] to eliminate the error caused by the command-filter, which inevitably increased the complexity of the control algorithm. Meanwhile, the study in [15] did not provide an effective method for handling external disturbances.

In many real-world engineering systems, disturbances are inevitable and often result in significant deterioration of control performance. This highlights the necessity of studying control problems in nonlinear systems subjected to external disturbances. To cope with this challenge, significant research efforts have been dedicated to addressing the associated control problem [16], [17], [18], [19], [20]. The study in [21] designed a disturbance observer to estimate the external disturbance for strict-feedback nonlinear systems. An adaptive sliding mode disturbance observer was developed in [22] to estimate and compensate for lumped disturbances, including model uncertainties and friction, thereby improving the robustness of fixed-time tracking control for uncertain robotic systems

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subject to actuator saturation. To mitigate the adverse effects of system uncertainties, environmental disturbances, and flexible vibrations during spacecraft maneuvers, a fixed-time disturbance observer (FxTDO) was introduced in [23] to estimate and compensate for the resulting lumped disturbances. However, the disturbance observer designed in [21], [22], and [23] only estimates the disturbance itself, without considering the estimation of its higher-order derivatives. In [24], a continuous finite-time disturbance observer (FTDO) was developed to estimate both the disturbances and their derivatives. However, the settling time of the disturbance observer still depended on the initial conditions of the system in [24]. As a result, developing an FxTDO is of significant research value. At the same time, event-triggered control has attracted significant research attention due to its potential for optimizing the use of limited network resources [25], [26], [27], [28], [29], [30], [31], [32]. In this context, dynamic event-triggered control has gained increasing popularity due to its enhanced adaptability and efficiency [33], [34], [35], [36], [37], [38]. Hence, it is crucial to implement a dynamic event-triggered control strategy to conserve network resources.

Based on the previous analysis, a practical fixed-time tracking control strategy based on the FxTDO is proposed. Within the standard Lyapunov-based analysis framework, the main contributions of this article are summarized as follows.

1) A FxTDO is developed to simultaneously estimate the unknown external disturbances and their higher-order derivatives. Through a sequence of coordinate transformations, the disturbance estimation task is reformulated as a fixed-time observation problem. The resulting estimation errors are guaranteed to converge to zero within a fixed time T_{do}, and the upper bound of the convergence time is independent of the initial conditions. Compared with the FTDO in [24], the proposed observer removes the dependence of the convergence time on the initial conditions and provides the disturbance information required for the subsequent controller design.
2) To avoid the repeated differentiation of virtual controllers in the backstepping procedure, a fixed-time command filter (FxTCF) is employed to alleviate the resulting computational complexity. The proposed filter guarantees that the filtering error converges to zero within a fixed time T_α. Consequently, the mismatch between the virtual controller and the output of the command filter is eliminated after T_α. Compared with the command-filters in [13], [14], and [15], the proposed filter prevents persistent filtering errors from adversely affecting the tracking performance and facilitates the establishment of practical fixed-time stability.
3) A nonsingular backstepping control framework with a command-filter is developed without introducing additional dynamics for error compensation. A switching function is incorporated into the virtual controllers to ensure the required second-order differentiability and the effective implementation of the FxTCF. In contrast to the methods in [13] and [15], which require an auxiliary mechanism to compensate for filtering errors, the proposed design simplifies the controller structure and guarantees practical fixed-time tracking for nonlinear systems in strict-feedback form under external disturbances.

The remainder of this article is outlined as follows: the system model, the control objective, and some preliminary works are proposed in Section II. In Section III, the FxTDO, FxTCF, and the control input under the dynamic event-triggered mechanism are designed. Simulation outcomes and conclusion are introduced in Sections IV and V, respectively.

II. Problem Description and Preliminaries

The class of strict-feedback nonlinear systems under consideration is described by

$$\begin{cases} \dot{x}_i = x_{i+1} + f_i(\bar{x}_i) + d_i, \quad i = 1, \ldots, n-1 \\ \dot{x}_n = u + f_n(\bar{x}_n) + d_n \\ y = x_1 \end{cases} \tag{1}$$

where $\bar{x}_i = [x_1, \ldots, x_i]^T \in \mathbb{R}^i$. The control input and output of the system are denoted by $u \in \mathbb{R}$ and $y \in \mathbb{R}$, respectively. The nonlinear functions $f_i(\bar{x}_i)$ are known to be bounded and smooth, while d_i represents unknown external disturbances. The desired reference trajectory is given by $y_r(t)$.

A. Control Objective

The primary goal pursued in this work is to formulate the control strategy and design the FxTDO for the nonlinear system (1) such that 1) the tracking error converges to a compact residual set within a fixed time, thereby ensuring practical fixed-time tracking of the reference signal y_r, 2) all the signals are guaranteed to remain bounded, and 3) the errors of observer converge to zero within a fixed time.

With the aim of accomplishing the aforementioned control objective, a definition, some fundamental assumptions, and lemmas are established to aid the following controller design and rigorous analysis.

Definition 1: $\lceil \omega \rfloor^l = sign(\omega)|\omega|^l$, $l > 0$ $\forall \omega \in \mathbb{R}$.

Assumption 1 [24]: The external disturbance $d_i(t)$ is considered to be produced by the following system:

$$\dot{\upsilon}_i = A_i \upsilon_i + B_i \varrho_i, \quad d_i = \upsilon_{i1} \tag{2}$$

where $\upsilon_i = [\upsilon_{i,1}, \upsilon_{i,2}, \ldots, \upsilon_{i,q}]^T \in \mathbb{R}^q$, ϱ_i denotes an unknown constant and the matrices A_i and B_i are specified as

$$A_i = \begin{bmatrix} 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ a_{i1} & a_{i2} & \cdots & a_{iq} \end{bmatrix}, \quad B_i = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \tag{3}$$

where $a_{i1}, \ldots, a_{iq}$ are known constants.

Remark 1: The disturbance model in ***Assumption 1***, though restrictive, is not merely theoretical. By selecting appropriate matrices A_i and B_i, this exosystem framework is capable of generating a wide range of disturbances commonly observed in engineering practice. For example, constant disturbances [39], ramp disturbances in robot systems [40], harmonic disturbances [41] in power systems, and the disturbances in the multidimensional heat equation [42]. These cases demonstrate

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that the adopted model covers a broad class of structured disturbances of practical relevance. Nevertheless, it should be emphasized that the proposed observer cannot handle unstructured or nonlinear disturbances that fall outside this class. Extending the framework to such general disturbances constitutes an important and challenging topic for future research.

Assumption 2 [35]: The desired reference trajectory $y_r(t)$, together with its nth order time derivatives are assumed to be continuous and bounded.

Lemma 1 [15]: For $\psi(t) \in \mathbb{R}^m$, the switching function takes the form

$$\Phi(\psi(t)) = \begin{cases} \dfrac{\psi}{\left(\psi^T\psi\right)^{1-p}}, & \psi^T\psi > k \\ sin^2\left(\dfrac{\psi^T\psi\pi}{2k}\right)\dfrac{\psi}{\left(\psi^T\psi\right)^{1-p}}, & \psi^T\psi \le k \end{cases} \tag{4}$$

where $1/2 < p < 1$ and k is a arbitrary small positive constant. If $\ddot{\psi}(t)$ exists, $\ddot{\Phi}(\psi(t))$ exists.

Lemma 2 [43]: The following scalar system is taken into account:

$$\dot{V}(x) \le -\xi_1 V^{\alpha_0}(x) - \xi_2 V^{\beta_0}(x) + \gamma \tag{5}$$

where $V(x)$ denotes a smooth positive definite function, with parameters $\xi_1 > 0$, $\xi_2 > 0$, $\alpha_0 > 1$, $0 < \beta_0 < 1$, and $0 < \gamma < \infty$. Under these conditions, the system achieves practical fixed-time stability, and the settling time can be bounded by

$$T \le \frac{1}{\xi_1 \nu(\alpha_0 - 1)} + \frac{1}{\xi_2 \nu(1 - \beta_0)} \tag{6}$$

where ν is a design parameter, which satisfies $0 < \nu < 1$. Moreover, for $0 < \rho < 1$, the solution of system (5) takes the following form:

$$\Omega_x = \left\{ x | V(x) \le min\left\{ \left(\frac{\gamma}{\xi_1(1-\rho)}\right)^{\frac{1}{\alpha_0}}, \left(\frac{\gamma}{\xi_2(1-\rho)}\right)^{\frac{1}{\beta_0}} \right\} \right\}. \tag{7}$$

Lemma 3 [44]: For $0 < p \le 1$, $\Theta_i \in \mathbb{R}$ and $\chi_i \ge 0$, it holds that

$$\left(\sum_{i=1}^{n} |\Theta_i|\right)^p \le \sum_{i=1}^{n} |\Theta_i|^p, \left(\sum_{i=1}^{n} \chi_i\right)^2 \le n \sum_{i=1}^{n} \chi_i^2. \tag{8}$$

Lemma 4 [45]: The following fixed-time observer is formulated:

$$\begin{cases} \dot{\vartheta}_1 = \vartheta_2 - k_1 \lceil e_0 \rfloor^{\lambda_1} - k_2 \lceil e_0 \rfloor^{\lambda_2} \\ \quad \vdots \\ \dot{\vartheta}_i = \vartheta_{i+1} - k_{2i-1} \lceil e_0 \rfloor^{\lambda_{2i-1}} - k_{2i} \lceil e_0 \rfloor^{\lambda_{2i}} \\ \quad \vdots \\ \dot{\vartheta}_n = -k_{2n-1} \lceil e_0 \rfloor^{\lambda_{2n-1}} - k_{2n} \lceil e_0 \rfloor^{\lambda_{2n}} \end{cases} \tag{9}$$

where e_0 denotes the observation error, defined as $e_0 = \vartheta_1 - \vartheta_0$, with ϑ_0 representing the ideal signal. The parameters $\lambda_{2i-1}(i = 1, 2, \ldots, n)$ are specified by $\lambda_{2i-1} = i\lambda_{01} - (i-1)$, where $\lambda_{01} \in (1-\epsilon, 1)$, where $\epsilon > 0$ is extremely small. The parameters $\lambda_{2i}(i = 1, 2, \ldots, n)$ are specified by $\lambda_{2i} = i\lambda_{02} - (i-1)$, where $\lambda_{02} \in (1, 1+\epsilon)$. By appropriately assigning the observer gains k_{2i-1} and k_{2i}, the associated matrices K_1 and K_2 are guaranteed to be Hurwitz, thereby ensuring that the observation error e_0 converges to zero within a fixed time T_0

$$K_1 = \begin{bmatrix} -k_1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -k_{2n-3} & 0 & \cdots & 1 \\ -k_{2n-1} & 0 & \cdots & 0 \end{bmatrix} \tag{10}$$

$$K_2 = \begin{bmatrix} -k_2 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -k_{2n-2} & 0 & \cdots & 1 \\ -k_{2n} & 0 & \cdots & 0 \end{bmatrix}. \tag{11}$$

Moreover, the convergence time T_o is given by

$$T_o \le \frac{\lambda_{max}^{1-\lambda_{01}}(X_1)}{\underline{\lambda}_1(1-\lambda_{01})} + \frac{1}{\underline{\lambda}_2(\lambda_{02}-1)\gamma_0^{\lambda_{02}-1}} \tag{12}$$

where $\underline{\lambda}_1 = \lambda_{min}(Y_1)/\lambda_{max}(X_1)$, $\underline{\lambda}_2 = \lambda_{min}(Y_2)/\lambda_{max}(X_2)$, and $0 < \gamma_0 < \lambda_{min}(X_2)$. The positive definite nonsingular matrices X_1, X_2, Y_1, and Y_2 satisfy the equations $X_1K_1 + K_1^TX_1 = -Y_1$ and $X_2K_2 + K_2^TX_2 = -Y_2$.

III. Main Results

A. Design of FxTDO and FxTCF

This subsection focuses on the design of the FxTDO and FxTCF. The FxTDO reconstructs the disturbance together with the corresponding higher-order derivative information in fixed time, whereas the FxTCF mitigates the explosion of complexity arising from repeated differentiation in the backstepping procedure.

As a preliminary step in constructing the FxTDO, define the following coordinate transformations:

$$\begin{cases} \eta_{i1} = x_i \\ \eta_{i2} = d_i - a_{iq}\eta_{i1} \\ \eta_{i3} = d_i^{(1)} - a_{iq}d_i - a_{i(q-1)}\eta_{i1} \\ \eta_{ij} = d_i^{(j-2)} - a_{iq}d_i^{(j-3)} - \cdots - a_{i(q-j+4)}d_i^{(1)} \\ \qquad - a_{i(q-j+3)}d_i - a_{i(q-j+2)}\eta_{i1}, j = 4, \ldots, q \\ \eta_{i(q+1)} = d_i^{(q-1)} - a_{iq}d_i^{(q-2)} - \cdots - a_{i2}d_i - a_{i1}\eta_{i1} \\ \eta_{i(q+2)} = \varrho_i. \end{cases} \tag{13}$$

By differentiating coordinate transformations (13), the dynamics of η_{ij} can be expressed as

$$\begin{cases} \dot{\eta}_{i1} = \eta_{i2} + \varsigma_{i1} \\ \quad \vdots \\ \dot{\eta}_{ij} = \eta_{i(j+1)} + \varsigma_{ij} \\ \quad \vdots \\ \dot{\eta}_{i(q+1)} = \eta_{i(q+2)} + \varsigma_{i(q+1)} \\ \dot{\eta}_{i(q+2)} = 0 \end{cases} \tag{14}$$

where $\varsigma_{i1} = a_{iq}\eta_{i1} + (f_i + x_{i+1})$, $\varsigma_{ij} = a_{i(q-j+1)}\eta_{i1} - a_{i(q-j+2)}(f_i + x_{i+1})(j = 2, \ldots, q)$ and $\varsigma_{i(q+1)} = -a_{i1}(f_i + x_{i+1})$.

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On this basis, the FxTDO is constructed as

$$\begin{cases} \dot{\hat{\eta}}_{i1} = \hat{\eta}_{i2} + \varsigma_{i1} + p_{i1}\lceil \varepsilon_{i1} \rfloor^{\alpha_{i1}} + q_{i1}\lceil \varepsilon_{i1} \rfloor^{\beta_{i1}} \\ \quad \vdots \\ \dot{\hat{\eta}}_{ij} = \hat{\eta}_{i(j+1)} + \varsigma_{ij} + p_{ij}\lceil \varepsilon_{i1} \rfloor^{\alpha_{ij}} + q_{ij}\lceil \varepsilon_{i1} \rfloor^{\beta_{ij}} \\ \quad \vdots \\ \dot{\hat{\eta}}_{i(q+2)} = p_{i(q+2)}\lceil \varepsilon_{i1} \rfloor^{\alpha_{i(q+2)}} + q_{i(q+2)}\lceil \varepsilon_{i1} \rfloor^{\beta_{i(q+2)}} \end{cases} \tag{15}$$

where $\varepsilon_{i1} = \eta_{i1} - \hat{\eta}_{i1}$. $0 < \alpha_{ij} < 1$ are defined as $\alpha_{ij} = j\alpha_0 - (j-1)$, where $\alpha_0 \in (1-\epsilon, 1)$ for an extremely small positive constant ϵ. The parameters $\beta_{ij}, j = 1, 2, \ldots, q+2$ are defined as $\beta_{ij} = j\beta_0 - (j-1)$, where $\beta_0 \in (1, 1+\epsilon)$. p_{ij} and q_{ij} represent the observer gains.

Consequently, it follows that the observation error system takes the form:

$$\begin{cases} \dot{\varepsilon}_{i1} = \varepsilon_{i2} - p_{i1}\lceil \varepsilon_{i1} \rfloor^{\alpha_{i1}} - q_{i1}\lceil \varepsilon_{i1} \rfloor^{\beta_{i1}} \\ \quad \vdots \\ \dot{\varepsilon}_{ij} = \varepsilon_{i(j+1)} - p_{ij}\lceil \varepsilon_{i1} \rfloor^{\alpha_{ij}} - q_{ij}\lceil \varepsilon_{i1} \rfloor^{\beta_{ij}} \\ \quad \vdots \\ \dot{\varepsilon}_{i(q+2)} = -p_{i(q+2)}\lceil \varepsilon_{i1} \rfloor^{\alpha_{i(q+2)}} - q_{i(q+2)}\lceil \varepsilon_{i1} \rfloor^{\beta_{i(q+2)}} \end{cases} \tag{16}$$

where $\varepsilon_{ij} = \eta_{ij} - \hat{\eta}_{ij}, j = 2, \ldots, q+2$.

Second, to overcome the challenge of "complexity explosion," design the subsequent FxTCF

$$\begin{cases} \dot{\bar{\alpha}}_i = \alpha_{id} - m_{i1}\lceil e_{\alpha_i} \rfloor^{\gamma_{i1}} - m_{i2}\lceil e_{\alpha_i} \rfloor^{\gamma_{i2}} \\ \dot{\alpha}_{id} = -m_{i3}\lceil e_{\alpha_i} \rfloor^{\gamma_{i3}} - m_{i4}\lceil e_{\alpha_i} \rfloor^{\gamma_{i4}} \end{cases} \tag{17}$$

where e_{α_i} is the filtering error, which satisfies $e_{\alpha_i} = \bar{\alpha}_i - \alpha_{i-1}$. $\bar{\alpha}_i$ is the output of the command filter, and α_{i-1} is the virtual controller, which will be designed in Section III-B. The exponents γ_{i1} and γ_{i3} satisfy the relations $\gamma_{i1} = \gamma_{01}$ and $\gamma_{i3} = 2\gamma_{01} - 1$, respectively. γ_{01} is a positive constant, which satisfies $\gamma_{01} \in (1-\epsilon, 1)$. The exponents γ_{i2} and γ_{i4} satisfy the relations $\gamma_{i2} = \gamma_{02}$ and $\gamma_{i4} = 2\gamma_{02} - 1$, respectively. γ_{02} is a positive constant, which satisfies $\gamma_{02} \in (1, 1+\epsilon)$. m_{i1}, m_{i2}, m_{i3}, and m_{i4} are the filter gains.

Theorem 1: Through a suitable choice of the observer gains p_{ij}, q_{ij}, and filter gains $m_{ik}(k = 1, \ldots, 4)$, the observation error ε_{ij} and filtering error e_{α_i} are ensured to converge to zero in fixed time T_{do} and T_α, respectively.

Proof: Under a suitable choice of p_{ij} and q_{ij}, the associated matrices satisfy the Hurwitz condition. It then follows from *Lemma 4* that the error vector $\varepsilon_i = [\varepsilon_{i1}, \ldots, \varepsilon_{i(q+2)}]^T$ vanishes within a fixed time T_{do}:

$$P = \begin{bmatrix} -p_{i1} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -p_{i(q+1)} & 0 & \cdots & 1 \\ -p_{i(q+2)} & 0 & \cdots & 0 \end{bmatrix} \tag{18}$$

$$Q = \begin{bmatrix} -q_{i1} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -q_{i(q+1)} & 0 & \cdots & 1 \\ -q_{i(q+2)} & 0 & \cdots & 0 \end{bmatrix}. \tag{19}$$

Likewise, it follows from *Lemma 4* that e_{α_i} vanishes within a fixed time T_α, under suitable selection of the filter gains that ensures the Hurwitz property of the matrices M_1 and M_2, which are defined as follows:

$$M_1 = \begin{bmatrix} -m_{i1} & 1 \\ -m_{i3} & 0 \end{bmatrix}, \quad M_2 = \begin{bmatrix} -m_{i2} & 1 \\ -m_{i4} & 0 \end{bmatrix}. \tag{20}$$

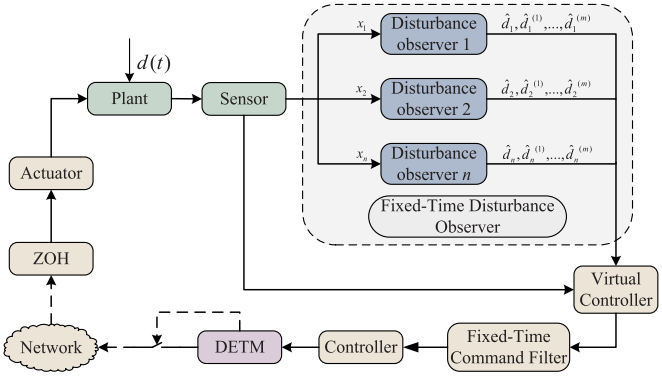


Fig. 1. Structural diagram of the proposed control framework.

Remark 2: The FxTDO formulated in (15) is developed based on the observer structure in [45], rather than being a direct application of the existing method. By introducing a sequence of coordinate transformations, the proposed observer is extended to estimate not only the disturbance itself but also its higher-order derivatives. Furthermore, unlike the FTDO in [24], the proposed observer guarantees that the associated estimation errors converge to zero within a fixed time T_{do} independent of the initial conditions. Therefore, the proposed observer provides enhanced disturbance-estimation capability and improves the disturbance rejection capability of the closed-loop system.

Remark 3: Unlike the conventional command-filter technique that requires an additional error compensation system to address the filtering error [15], the FxTCF in (17) guarantees that the filtering error converges to zero within a fixed time T_α. The observer gains p_{ij} and q_{ij}, together with the filter gains m_{ik}, are selected such that the matrices P, Q, M_1, and M_2 are Hurwitz. From a control perspective, these gains mainly affect the convergence rate, control effort, and robustness margin of the closed-loop system. Larger gains generally accelerate the convergence process, but may also require greater control action and amplify the influence of measurement noise. Therefore, the gain selection should balance convergence speed, control performance, and implementation robustness.

B. Design of Control Input for the Controlled System

This section develops a controller through the backstepping framework, where the FxTCF, introduced previously, serves to overcome the complexity in the design process. In addition, to facilitate understanding of the designed control framework, its structural diagram is provided in Fig. 1.

To begin with, the system errors are constructed in the following form:

$$\begin{cases} z_1 = x_1 - y_r \\ z_i = x_i - \bar{\alpha}_i, \quad i = 2, \ldots, n. \end{cases} \tag{21}$$

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Step 1: Select the Lyapunov candidate function as

$$V_1 = \frac{1}{2} z_1^T z_1. \tag{22}$$

Accordingly, differentiating V_1 gives

$$\dot{V}_1 = z_1 \left(x_2 + f_1 + d_1 - \dot{y}_r \right). \tag{23}$$

Next, the virtual controller α_1 is constructed in the form

$$\alpha_1 = -h_1 z_1 \left(z_1^T z_1 \right) - r_1 \Phi \left(z_1 \right) - f_1 - \hat{d}_1 + \dot{y}_r \tag{24}$$

where $h_1 > 0$ and $r_1 > 0$ are user specified constants. Using (24) in (23) yields

$$\begin{aligned} \dot{V}_1 = & -h_1 \left(z_1^T z_1 \right)^2 - r_1 z_1 \Phi \left(z_1 \right) + z_1 z_2 + z_1 \tilde{d}_1 \\ & + z_1 \left(\bar{\alpha}_2 - \alpha_1 \right). \end{aligned} \tag{25}$$

Step 2: Select $V_2 = V_1 + 1/2 z_2^T z_2$ as the Lyapunov function. Combining this definition with (25) gives

$$\begin{aligned} \dot{V}_2 = & -h_1 \left(z_1^T z_1 \right)^2 - r_1 z_1 \Phi \left(z_1 \right) + z_2 z_3 + \sum_{j=1}^{2} z_j \left(\bar{\alpha}_{j+1} - \alpha_j \right) \\ & + \sum_{j=1}^{2} z_j \tilde{d}_j + z_2 \left(z_1 + \alpha_2 + f_2 + \hat{d}_2 - \dot{\bar{\alpha}}_2 \right). \end{aligned} \tag{26}$$

The virtual controller α_2 is constructed in the form

$$\alpha_2 = -h_2 z_2 \left(z_2^T z_2 \right) - r_2 \Phi \left(z_2 \right) - z_1 - f_2 - \hat{d}_2 + \dot{\bar{\alpha}}_2 \tag{27}$$

where h_2 and r_2 denote the positive design parameters.

Based on (27), the following expression is derived:

$$\begin{aligned} \dot{V}_2 = & -\sum_{j=1}^{2} h_j \left(z_j^T z_j \right)^2 - \sum_{j=1}^{2} r_j z_j \Phi \left(z_j \right) + z_2 z_3 \\ & + \sum_{j=1}^{2} z_j \left(\bar{\alpha}_{j+1} - \alpha_j \right) + \sum_{j=1}^{2} z_j \tilde{d}_j. \end{aligned} \tag{28}$$

Step i (3 ≤ i ≤ n-1): Choose the following Lyapunov function:

$$V_i = V_{i-1} + \frac{1}{2} z_i^T z_i. \tag{29}$$

Further, $\dot{V}_i$ is calculated as

$$\begin{aligned} \dot{V}_i = & -\sum_{j=1}^{i-1} h_j \left(z_j^T z_j \right)^2 - \sum_{j=1}^{i-1} r_j z_j \Phi \left(z_j \right) + z_{i-1} z_i + \sum_{j=1}^{i-1} z_j \tilde{d}_j \\ & + \sum_{j=1}^{i-1} z_j \left(\bar{\alpha}_{j+1} - \alpha_j \right) + z_i \left(\alpha_i + f_i + \hat{d}_i - \dot{\bar{\alpha}}_i \right) \\ & + z_i z_{i+1} + z_i \tilde{d}_i + z_i \left(\bar{\alpha}_{i+1} - \alpha_i \right). \end{aligned} \tag{30}$$

Based on the preceding analysis, the virtual control law takes the form

$$\alpha_i = -h_i z_i \left(z_i^T z_i \right) - r_i \Phi \left(z_i \right) - z_{i-1} - f_i - \hat{d}_i + \dot{\bar{\alpha}}_i \tag{31}$$

where h_i and r_i denote positive design parameters.

Upon substitution of (31) into (30), the subsequent result is obtained

$$\begin{aligned} \dot{V}_i = & -\sum_{j=1}^{i} h_j \left(z_j^T z_j \right)^2 - \sum_{j=1}^{i} r_j z_j \Phi \left(z_j \right) + z_i z_{i+1} \\ & + \sum_{j=1}^{i} z_j \tilde{d}_j + \sum_{j=1}^{i} z_j \left(\bar{\alpha}_{j+1} - \alpha_j \right). \end{aligned} \tag{32}$$

Step n: Following the previous steps, V_n is introduced in the following form:

$$V_n = V_{n-1} + \frac{1}{2} z_n^T z_n. \tag{33}$$

By following the procedures in (30)–(32), the subsequent equation is derived:

$$\begin{aligned} \dot{V}_n = & -\sum_{j=1}^{n-1} h_j \left(z_j^T z_j \right)^2 - \sum_{j=1}^{n-1} r_j z_j \Phi \left(z_j \right) + \sum_{j=1}^{n} z_j \tilde{d}_j \\ & + \sum_{j=1}^{n-1} z_j \left(\bar{\alpha}_{j+1} - \alpha_j \right) + z_n \left(u - \alpha_n \right) \\ & + z_n \left(z_{n-1} + \alpha_n + f_n + \hat{d}_n - \dot{\bar{\alpha}}_n \right). \end{aligned} \tag{34}$$

To complete the recursive design, α_n is designed as

$$\alpha_n = -h_n z_n \left(z_n^T z_n \right) - r_n \Phi \left(z_n \right) - z_{n-1} - f_n - \hat{d}_n + \dot{\bar{\alpha}}_n \tag{35}$$

where h_n and r_n denote positive design parameters.

Next, $\dot{V}_n$ is calculated as

$$\begin{aligned} \dot{V}_n = & -\sum_{j=1}^{n} h_j \left(z_j^T z_j \right)^2 - \sum_{j=1}^{n} r_j z_j \Phi \left(z_j \right) + \sum_{j=1}^{n} z_j \tilde{d}_j \\ & + \sum_{j=1}^{n-1} z_j \left(\bar{\alpha}_{j+1} - \alpha_j \right) + z_n \left(u - \alpha_n \right). \end{aligned} \tag{36}$$

To alleviate communication burden, a dynamic event-triggered mechanism is introduced, leading to the following controller:

$$\begin{aligned} v \left(t \right) &= \alpha_n - \left(b_0 + \frac{\theta}{a_0} \right) sign \left(z_n \right) \\ u \left(t \right) &= v \left(t_k \right), \quad t \in \left[t_k, t_{k+1} \right) \end{aligned} \tag{37}$$

where a_0, b_0 are positive constants.

The condition for dynamic event-triggered is formulated as

$$t_{k+1} = inf \left\{ t > t_k \middle| \theta + a_0 \left(b_0 - \left| e \left(t \right) \right| \right) \le 0 \right\} \tag{38}$$

where $e(t) = u(t) - v(t)$. t_k is the current triggering moment and t_{k+1} is the next triggering moment. The dynamic variable θ can be obtained from the following dynamic expression:

$$\dot{\theta} = -\iota \theta + a_0 \left(b_0 - \left| e \left(t \right) \right| \right). \tag{39}$$

Lemma 5 [46]: Considering the condition for dynamic event-triggered (38) and dynamic expression (39), the dynamic variable θ satisfies $0 \le \theta \le a_0 b_0 / \iota$.

C. Stability Analysis

To confirm the stability of the system, the following theorem is formulated:

Theorem 2: For the strict-feedback nonlinear system (1) subject to external disturbances, the application of controller (37) ensures that

1) all system signals remain bounded;

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2) the output y converges to the desired reference trajectory $y_r(t)$ within fixed time T_0; and
3) the Zeno behavior is avoided.

Proof: Based on the FxTDO (15) and FxTCF (17), it can be concluded that the terms $\sum_{j=1}^{n} z_j \tilde{d}_j$ and $\sum_{j=1}^{n-1} z_j(\bar{\alpha}_{j+1} - \alpha_j)$ in (36) will converge to zero when the time $t \geq T_1$, where $T_1 = max\{T_{do}, T_\alpha\}$.

For the last term of (36), a time-varying function $\mu(t)$ can be defined such that $\mu(t_k) = 0$, $\mu(t_{k+1}) = \pm 1$ and $|\mu(t)| \leq 1$ for $\forall t \in [t_k, t_{k+1})$, which yields $v(t) = u(t) + (b_0 + \theta/a_0)\mu(t)$. Thus, it can be obtained that

$$\begin{aligned} z_n(u - \alpha_n) &= z_n\left(v(t) - \left(b_0 + \frac{\theta}{a_0}\right)\mu(t) - \alpha_n\right) \\ &= z_n\left(b_0 + \frac{\theta}{a_0}\right)(-sign(z_n) - \mu(t)). \end{aligned} \tag{40}$$

According to *Lemma 5*, it follows that $\theta > 0$ always holds. If $z_n \geq 0$, then $-sign(z_n) - \mu(t) \leq 0$ holds at all times, while if $z_n \leq 0$, then $-sign(z_n) - \mu(t) \geq 0$ consistently holds. Consequently, one can easily derive that $z_n(u - \alpha_n) \leq 0$.

As a result, (36) can be rewritten in the simplified form shown below

$$\dot{V}_n \leq -\sum_{j=1}^{n} h_j \left(z_j^T z_j\right)^2 - \sum_{j=1}^{n} r_j z_j \Phi(z_j). \tag{41}$$

The subsequent analysis considers three different cases of the switching function $\Phi(z_j)$, $j = 1, 2, \ldots, n$.

Case 1: When $z_j^T z_j > k$ for $\forall j = 1, 2, \ldots, n$, $\Phi(z_j) = z_j/((z_j^T z_j)^{1-p})$.

Under this condition, (41) can be computed as

$$\dot{V}_n = -\sum_{j=1}^{n} h_j \left(z_j^T z_j\right)^2 - \sum_{j=1}^{n} r_j \left(z_j^T z_j\right)^p. \tag{42}$$

Case 2: When $z_j^T z_j > k$ for some $j = 1, 2, \ldots, n$, $\Phi(z_j) = z_j/(z_j^T z_j)^{1-p}$. In this analysis, $z_j^T z_j \leq k$ is assumed to be valid solely for $j = 1$, i.e., $\Phi(z_j) = z_j/((z_j^T z_j)^{1-p})(j = 2, 3, \ldots, n)$.

It is straightforward to verify that $z_1\Phi(z_1) \geq (z_1^T z_1)^p - k^p$ holds under the condition $z_1^T z_1 \leq k$.

Thus, combining the preceding results gives

$$\begin{aligned} \dot{V}_n &= -\sum_{j=1}^{n} h_j \left(z_j^T z_j\right)^2 - r_1 z_1 \Phi(z_1) - \sum_{j=2}^{n} r_j \left(z_j^T z_j\right)^p \\ &\leq -\sum_{j=1}^{n} h_j \left(z_j^T z_j\right)^2 - \sum_{j=1}^{n} r_j \left(z_j^T z_j\right)^p + r_1 k^p. \end{aligned} \tag{43}$$

Case 3: When $z_j^T z_j \leq k$ for $\forall j = 1, 2, \ldots, n$, $\Phi(z_j) = sin^2(z_j^T z_j \pi/2k) z_j/(z_j^T z_j)^{1-p}$.

Similarly, one can conclude

$$\begin{aligned} \dot{V}_n &= -\sum_{j=1}^{n} h_j \left(z_j^T z_j\right)^2 - \sum_{j=1}^{n} r_j z_j \Phi(z_j) \\ &\leq -\sum_{j=1}^{n} h_j \left(z_j^T z_j\right)^2 - \sum_{j=1}^{n} r_j \left(z_j^T z_j\right)^p + \sum_{j=1}^{n-1} r_j k^p. \end{aligned} \tag{44}$$

Considering the results from all three cases, it is not difficult to conclude that (44) always holds. Hence, by applying *Lemma 3*, the subsequent result is derived

$$\begin{aligned} \dot{V}_n &= -\sum_{j=1}^{n} 4h_j \left(\frac{1}{2} z_j^T z_j\right)^2 - \sum_{j=1}^{n} 2^p r_j \left(\frac{1}{2} z_j^T z_j\right)^p + \sum_{j=1}^{n-1} r_j k^p \\ &\leq -\zeta V^2 - \xi V^p + \Lambda \end{aligned} \tag{45}$$

where $\Lambda = \sum_{j=1}^{n-1} r_j k^p$, $\zeta = min\{4h_1, 4h_2, \ldots, 4h_n\}/n$ and $\xi = min\{2^p r_1, 2^p r_2, \ldots, 2^p r_n\}$.

As indicated by *Lemma 2*, the settling time T_2 can be determined by

$$T_2 \leq \frac{1}{\zeta o_\tau} + \frac{1}{\xi o_\tau (1-p)} \tag{46}$$

where o_τ is defined as a positive constant, which satisfies $0 < o_\tau < 1$.

Moreover, the system errors $z_i (i = 1, 2, \ldots, n)$ is bounded by a compact residual set

$$\Omega_{z_i} = \left\{ z_i \left| V(z_i) \leq min\left\{ \left(\frac{\Lambda}{(1-o_\tau)\zeta}\right)^{\frac{1}{2}}, \left(\frac{\Lambda}{(1-o_\tau)\xi}\right)^{\frac{1}{p}} \right\} \right\}. \tag{47}$$

Given that the desired reference trajectory $y_r(t)$ is bounded, x_1 is guaranteed to remain bounded. The boundedness of the virtual controller α_1 can be inferred from (24). Due to the bounded input of (17), $\bar{\alpha}_2$ is also bounded. This implies that x_2 and α_2 are bounded. Repeating this analysis through all subsequent steps up to n confirms that the boundedness of x_j, α_j, and $\bar{\alpha}_j (j = 3, 4, \ldots, n)$ is ensured. According to *Lemma 5*, one can conclude that the dynamic variable θ is bounded, which implies that the event-triggered controller $u(t)$ is also bounded. The previous analysis leads to the conclusion that all signals of system (1) remain bounded.

The preceding analysis shows that the tracking error z_1 converges to a compact residual set Ω_{z_1} within a fixed time T_0, defined as

$$T_0 \leq T_1 + T_2. \tag{48}$$

Consequently, the validity of (i) and (ii) is established.

Next, the absence of Zeno behavior is verified in the following proof. First, we assume that the time interval $\Delta t = t_{k+1} - t_k$ converges to zero and $\Delta v = v(t_{k+1}) - v(t_k)$. Because of the continuity of α_n, $\lim_{\Delta t \to 0} \alpha_n(t_{k+1}) - \alpha_n(t_k) = 0$. Then, one can obtain that

$$|\Delta v| = \left(b_0 + \frac{\theta}{a_0}\right)\left|sign(z_n(t_k)) - sign\left(z_n\left(t_{k+1}\right)\right)\right|. \tag{49}$$

If $z_n = 0$, the continuity of z_n implies that at most one of its values at t_k and t_{k+1} can vanish, yielding $|sign(z_n(t_k)) - sign(z_n(t_{k+1}))| = 1$.

Conversely, if $z_n \neq 0$, its values at t_k and t_{k+1} share the same sign, so that $|sign(z_n(t_k)) - sign(z_n(t_{k+1}))| = 0$.

Combining these two cases yields

$$\lim_{\Delta t \to 0} |\Delta v| \leq \left(b_0 + \frac{\theta}{a_0}\right). \tag{50}$$

It is evident that (50) violates the event-triggered condition (38), which implies that $\Delta t \nrightarrow 0$. Hence, no Zeno behavior occurs.

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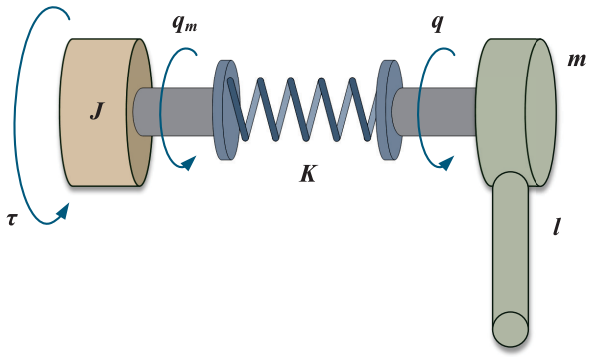


Fig. 2. Schematic of the flexible robotic manipulator.

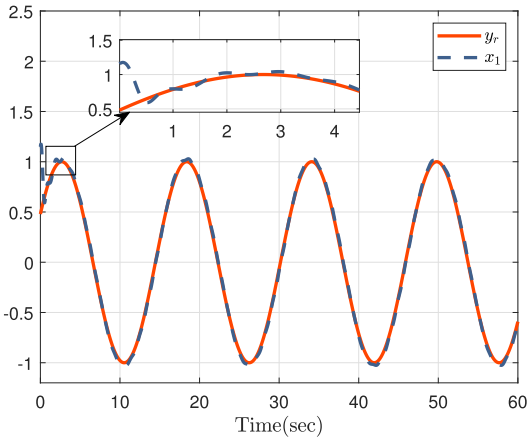


Fig. 3. Tracking performance of the controlled system.

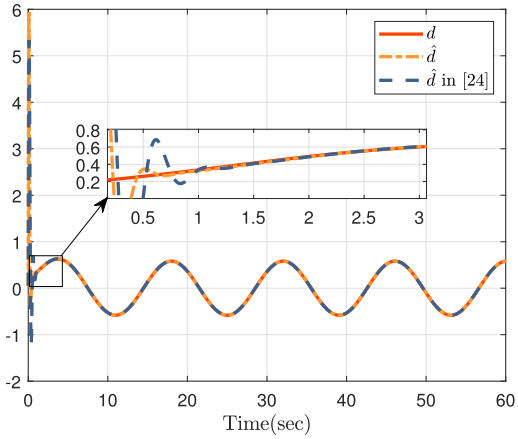


Fig. 4. Observation performance of d and its comparison with [24].

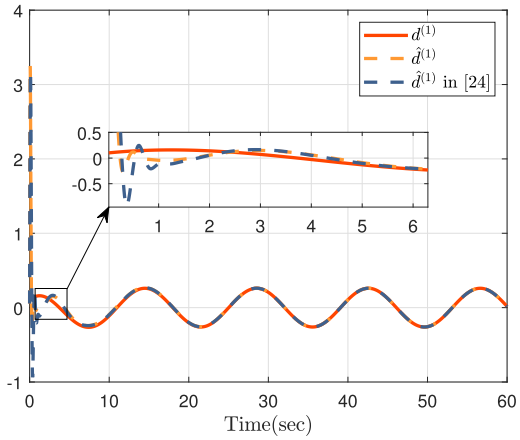


Fig. 5. Observation performance of $\dot{d}$ and its comparison with [24].

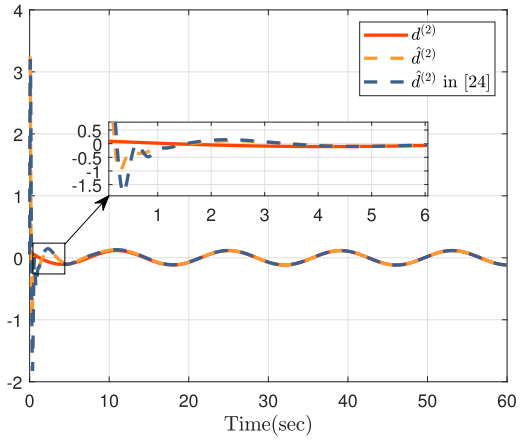


Fig. 6. Observation performance of $\ddot{d}$ and its comparison with [24].

IV. Simulation Example

To evaluate the performance of the designed controller, a simulation study is conducted on a flexible robotic manipulator, as illustrated in Fig. 2, with the corresponding system model given below:

$$\begin{cases} ml^2\ddot{q} + mgl\sin q + \cos\dot{q} = K(q_m - q) + d \\ J\ddot{q}_m + B\dot{q}_m + K(q_m - q) = \tau \\ y = q \end{cases} \tag{51}$$

where q, $\dot{q}$, and $\ddot{q}$ denote the link-side position, velocity, and acceleration vectors, whereas q_m, $\dot{q}_m$, and $\ddot{q}_m$ are their rotor-side counterparts. The parameters m, g, and l refer to the link mass, gravitational acceleration, and link length; K, J, and B characterize the joint flexibility, actuator inertia, and inherent damping. The input torque and external disturbance are denoted by τ and d, respectively. The desired reference trajectory is specified as $y_r = \sin(0.4t + 0.5)$.

Introduce the state variables $x_1 = q$, $x_2 = \dot{q}$, $x_3 = q_m$, and $x_4 = \dot{q}_m$, and set $u = \tau$. The resulting state-space model is

Remark 4: A tighter ultimate bound on z_i can be achieved by increasing h_j, or decreasing k and o_τ. Nevertheless, since k serves as the switching point of the piecewise function, choosing a small value of k inevitably amplifies the virtual controller, which may in turn lead to potential singularity issues. Meanwhile, if o_τ is selected too small, the upper bound of the settling time T_2 will become excessively large. Therefore, in practice, the selection of k requires a trade-off between reducing the steady-state error and maintaining the overall system performance.

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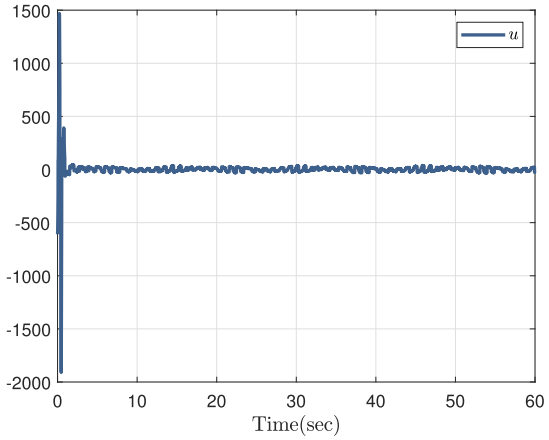


Fig. 7. Trajectory of the control input u.

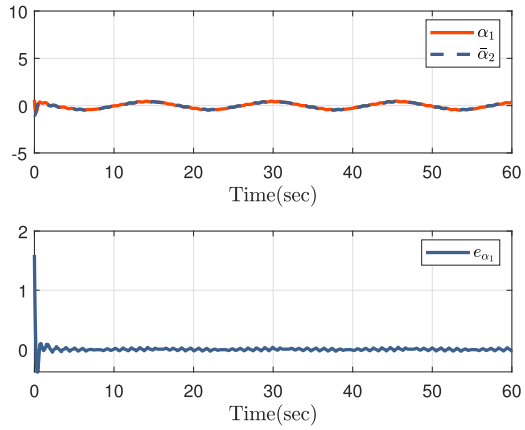


Fig. 8. Filtering performance and estimation error of the command-filter $\bar{\alpha}_2$.

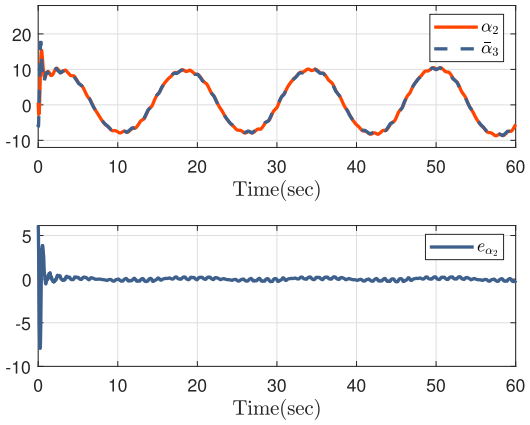


Fig. 9. Filtering performance and estimation error of the command-filter $\bar{\alpha}_3$.

given by

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \dfrac{K}{ml^2}(x_3 - x_1) - \dfrac{1}{ml^2}(mgl\sin x_1 + \cos x_2) + \dfrac{1}{ml^2}d \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = \dfrac{1}{J}u - \dfrac{B}{J}x_4 - \dfrac{K}{J}(x_3 - x_1) \end{cases} \tag{52}$$

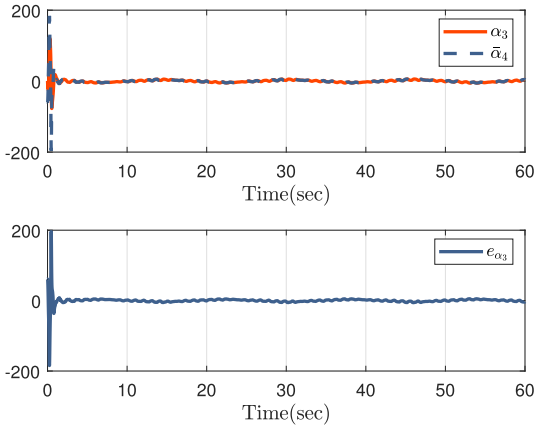


Fig. 10. Filtering performance and estimation error of the command-filter $\bar{\alpha}_4$.

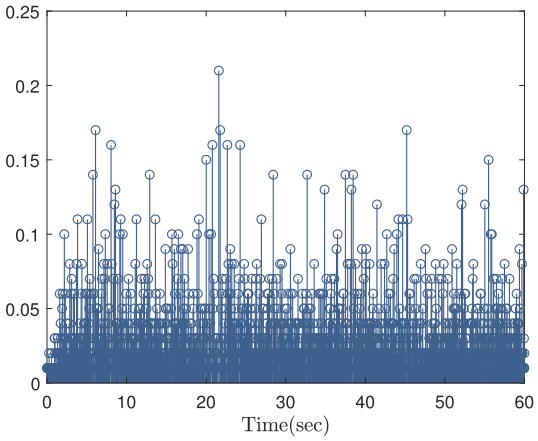


Fig. 11. Triggering times of dynamic event-triggered mechanism.

where d is described as

$$\begin{cases} d = \upsilon_{11} \\ \dot{\upsilon}_{11} = \upsilon_{12} \\ \dot{\upsilon}_{12} = \upsilon_{13} \\ \dot{\upsilon}_{13} = a_{11}\upsilon_{11} + a_{12}\upsilon_{12} + a_{13}\upsilon_{13} + \varrho_1. \end{cases} \tag{53}$$

The initial values of system states are selected as $x_0 = [1, 15, 0.65, 0.25, 0.25]$. The parameters of the system are specified as $m = 1$, $g = 9.8$, $l = 1$, $K = 1$, $J = 1$, and $B = 0.1$. The parameters of (53) are specified as $a_{11} = -0.1$, $a_{12} = -0.4$, $a_{13} = -0.6$, and $\varrho_1 = 0$.

Figs. 3–12 depict the outcomes of the simulation for system (52). Fig. 3 illustrates the tracking performance of the system. Figs. 4–6 describe the observation performance of d, $\dot{d}$ and $\ddot{d}$, together with a comparison against the method in [24]. The trajectories of the control input u is presented in Fig. 7. Figs. 8–10 describe the filtering performance of the designed FxTCF. Fig. 11 illustrates the triggering times of the dynamic event-triggered strategy. Fig. 12 presents a comparison of the observation times achieved by the proposed FxTDO and FTDO from [24], considering different initial error values. The comparison reveals that the proposed FxTDO exhibits only minor variations under different initial conditions.

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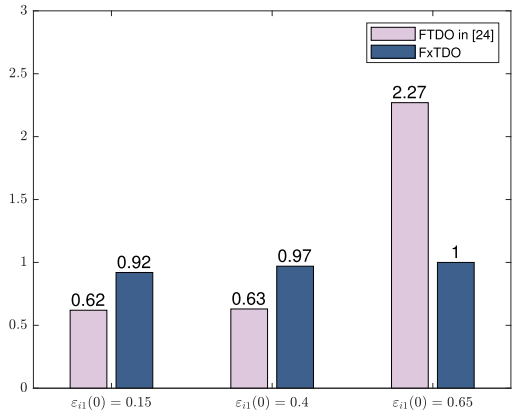


Fig. 12. Observation time comparison between the FTDO in [24] and the proposed FxTDO under different initial errors.

As evidenced by the simulation results, the designed control scheme provides great control performance and successfully fulfills the desired control objectives.

V. Conclusion

In this article, the dynamic event-triggered practical fixed-time tracking control strategy is proposed for nonlinear systems subject to external disturbances. The designed FxTDO enables precise estimation of disturbances and their high-order derivatives. To address the issue of "complexity explosion," the FxTCF is adopted, and the estimation error of the command-filter converges to zero within the fixed time. Moreover, the unnecessary use of communication bandwidth is reduced through the introduced dynamic event-triggered mechanism. The rationality and performance of the control strategy are substantiated through simulation analysis.

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